

Data Science Fundamentals [SC3021/SC5003]

Chapter 2: Data Science Thinking vs Computational Thinking

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Course content



Data science overview

What is Data Science?

Data Science Thinking vs. Computational Thinking

- Computational Thinking
- Data Science Thinking
- Comparison

Data Science Ecosystem



Data preparation and data management

Data profiling

Structuring

Enriching

Data cleaning

Data management platforms

Schema design

Querying data



Data analytics and understanding

Descriptive analytics

Diagnostic Analytics

Prescriptive analytics

Predictive analytics

Data visualization



Design considerations

Privacy

Security

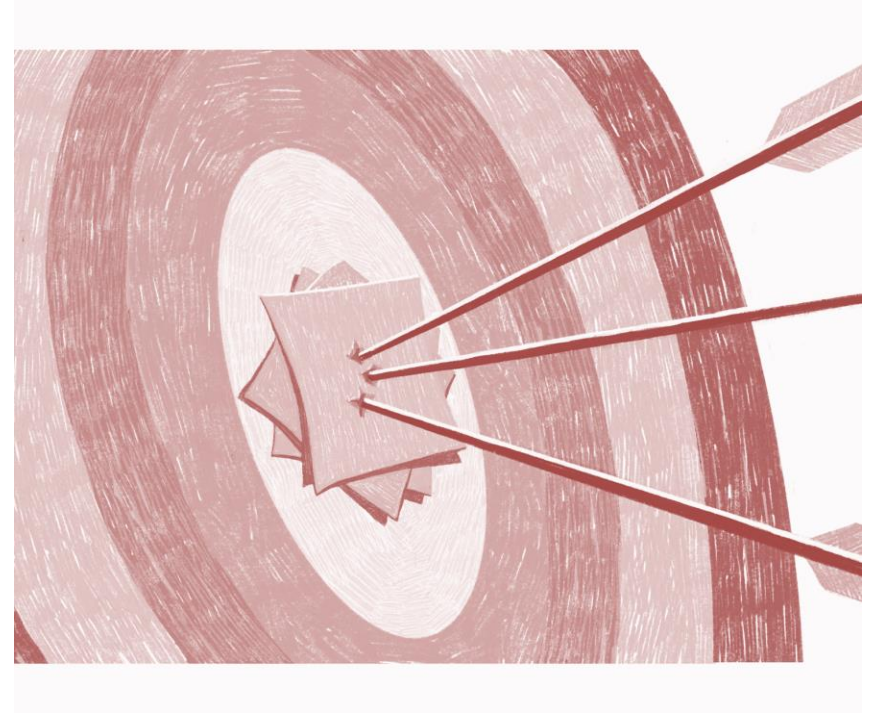
Ethics

Psychology

Policies

The content of this lecture is designed to enable you to...

- ...describe and compare the main characteristics of Computational Thinking and Data Science Thinking.
- ... explain how cross-functional thinking skills complement each other to solve a data science problem.
- ... argue why and how a problem should be addressed through Data Science Thinking (or not).
- ... conceive a rough agenda to solve a new problem by following the Data Science Thinking process.



COMPUTATIONAL THINKING



Computational Thinking introduction



- Computational Thinking is a **process** for solving **complex problems**.
- Thought process involved in **formulating a problem** and **expressing its solution** such that a computer, human, or machine can **effectively carry it out**.

Overview of key components



Decomposition



Pattern
recognition



Abstraction



Algorithm
design

Running (toy) example

Problem:
Preparing a kaya toast





Decomposition

- Break down a complex problem into subproblems.
 - Subproblems are smaller, more manageable
 - Subproblems are easier to solve
- Combining the solutions to all subproblems solves the overall problem.



Get bread



Get kaya



Get butter



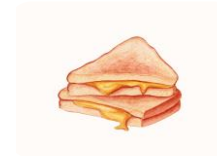
Toast bread



Spread kaya on one slice



Spread butter on the other slice



Put slices together



Pattern recognition

- Identify patterns or trends within a problem.
 - Steps following the same pattern may reuse same solution.
 - Patterns and trends allow to simplify the solution.



Get bread



Get kaya



Get butter



Toast bread



Spread kaya on one slice



Spread butter on the other slice



Put slices together

- Repeating pattern of spreading a filling (butter or kaya) on a slice.
- The pattern is repeated twice.



Abstraction

- Identify similarities and differences among similar problems to work toward a generic solution.
- Focuses on essential properties of a problem (invariant for similar problems)
- Ignores details of different problem instances.
- Simplifies problem solving by focusing on the fundamental characteristics of the type of problem to solve.

Similar problems 1: making different kinds of kaya toast

- Ignore specific brand of kaya, toast, butter, etc.
- Ignore specific type of toast
- Ignore specific type of toasting



Similar problems 2: making different types of simple sandwiches

- Ignore specific type of bread, spread
- All require three ingredients
All follow the same pattern of spreading a topping on a base and stacking these.
- Ignore (outsource) uncommon steps like toasting the bread before use.





Algorithm design

- Algorithm: a step-by-step plan to solve a problem.
- Can be used to solve similar problems as well.
- Previous Computational Thinking steps allow to define an algorithm at an appropriate generalization and abstraction level.

```
MakeSandwich(slice1, slice2, spread1, spread2){  
    half1 := Apply spread1 to one side of  
    slice1;  
    half2 := Apply spread2 to one side of  
    slice2;  
    Stack half1 and half2 s.t. sides with  
    spread face each other;  
}  
  
MakeKayaToast(){  
    initialize toast1, toast2, kaya, butter;  
    slice1 := toast(toast1);  
    slice2 := toast(toast2);  
    makeSandwich(slice1, slice2, kaya, butter);  
}
```

Computational Thinking (CT) and Computer Science (CS)

CT is applied within CS to create efficient algorithms and solutions to complex problems

CT during programming, data structures, algorithms, computer architecture, software engineering, etc.

CT CS

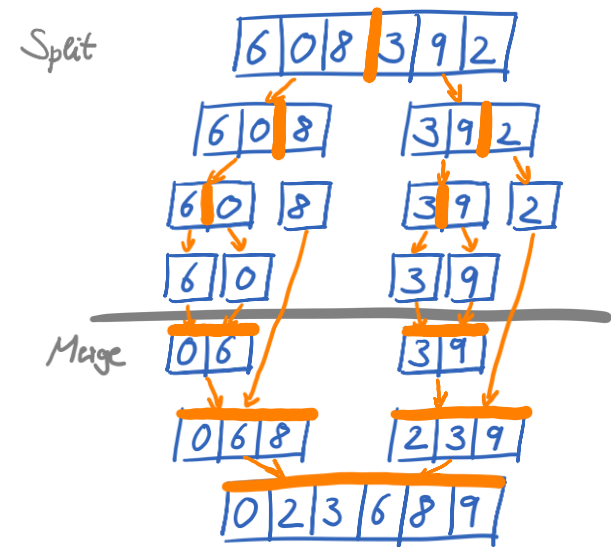
Modular programming (decomposition), data mining (pattern recognition), modeling (abstraction), algorithm analysis (algorithm design)

CS contributes tools and techniques to implement the CT process

CS example 1: Merge Sort

Problem: sort array [6,0,8,3,9,2] in ascending order

- **Decomposition:** The problem of sorting large array breaks down into smaller problems (i.e., sorting smaller arrays). The smallest problem size is one, leading to trivially sorted arrays.
- **Pattern Recognition:** Sorted subarrays can be merged efficiently to create larger sorted arrays.
- **Abstraction:** Specific values in the array are not relevant.
- **Algorithm Design:** Recursive design, dividing into a split phase (decomposition) and merge phase (based on recognized pattern).



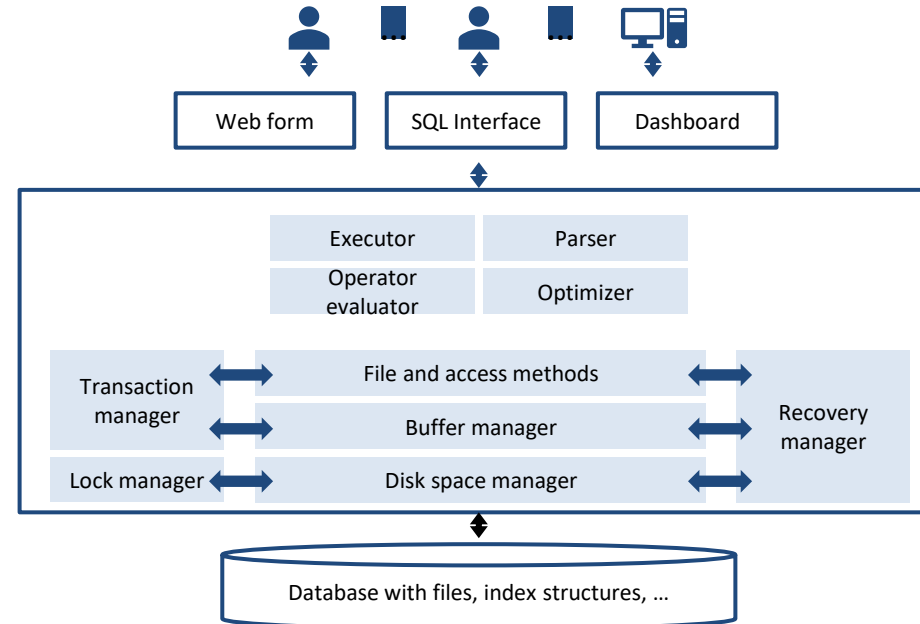
```
algorithm MergeSort ( $F$ )  $\rightarrow F_s$ 
  Input: an array to be sorted  $F$ 
  Output: a sorted array  $F_s$ 

  if  $F$  contains one element
  then return  $F$ 
  else
    Divide  $F$  into  $F_1$  and  $F_2$ ;
     $F_1 := \text{MergeSort}(F_1)$ ;
     $F_2 := \text{MergeSort}(F_2)$ ;
    return Merge( $F_1, F_2$ );
```

CS example 2: Database management systems

Problem: Support use of large amounts of data

- **Decomposition:** Break down the task into subtasks like data storage, retrieval, query processing, ...
- **Pattern recognition:** Patterns in data access are used to optimize query performance. Patterns in typical information need guide query language design...
- **Abstraction:** Table-based data model hides physical implementation details. Declarative query language focuses on what data is requested, not how to get it.
- **Algorithm design:** Numerous algorithms for efficient data storage, retrieval, query processing, ...



CS example 3: Autonomous vehicles

Problem: Build software for self-driving cars

1. **Decomposition:** Problem divides into subproblems such as perception, prediction, planning, control.
2. **Pattern recognition:** Use of machine learning algorithms to recognize patterns in sensor data to identify important features (e.g., traffic lights, pedestrians, other vehicles)
3. **Abstraction:** Focus is on essential features of the driving environment, Details irrelevant to autonomous driving are ignored (e.g., color of other cars).
4. **Algorithm design:** Development and use of various algorithms for different subtasks (e.g., search algorithms, control algorithms, machine learning algorithms)



Summary of Computational Thinking



- Computational Thinking is a **thought process** for solving **complex problems**.
- Allows to **express a problem solution** such that a computer, human, or machine can **effectively carry it out**.
- Process divides into **four main parts**, namely decomposition, pattern recognition, abstraction, and algorithm design.

DATA SCIENCE THINKING



Data Science Thinking introduction

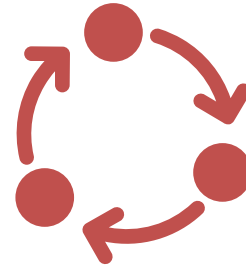


- Data Science Thinking is a **process** for solving **data science problems**.
 - Data science problems qualify as complex problems to be solved (cf. Computational Thinking)
 - The multi-disciplinary nature of data science problems (cf. definition of data science) requires methods of thinking beyond Computational Thinking.
- The notion of Data Science Thinking is quite new, with **many variations of its definition**.
- Goal of this lecture is to **highlight core aspects** that are reflected in most definitions.

Core concepts



Data Science Thinking **combines and adapts existing ways of thinking** to answer data science questions.

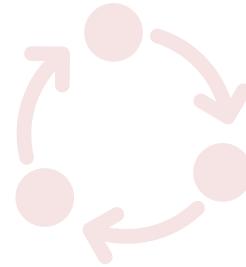


The process underlying Data Science Thinking is composed of **abstract steps to solve a problem of interest** to some stakeholder.

Core concepts



Data Science Thinking **combines and adapts existing ways of thinking** to answer data science questions.

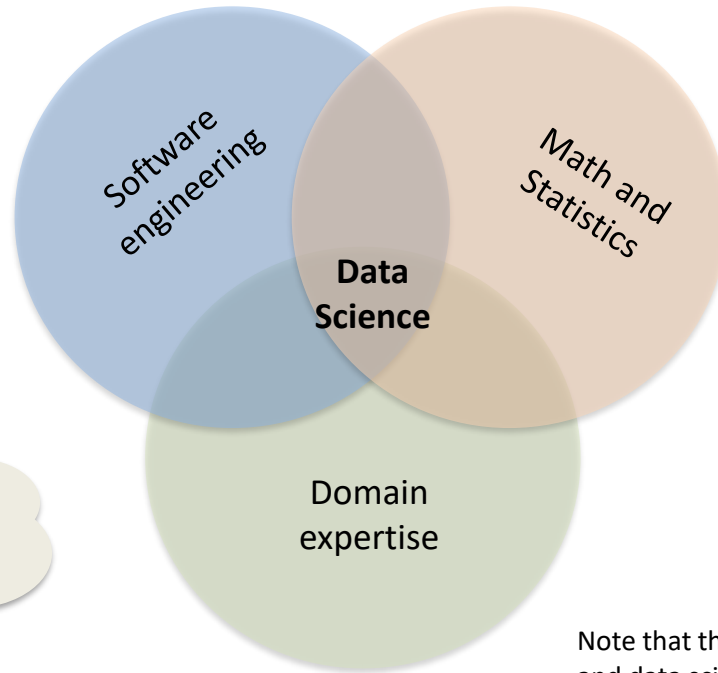


The **process** underlying Data Science Thinking is composed of **abstract steps to solve a problem of interest** to some stakeholder.



The different hats of data scientists

Think like an
engineer



Think like a
scientist

Think like a
stakeholder
(impact)

Note that there is no direct connection between ways of thinking and data science pillars, as the correspondence is not one-to-one.



Think like an engineer

- Why?
 - Need for code
 - Complex systems
 - Deploy models
 - Create systems to solve problems
- How?
 - Clean software engineering
 - Design for scalability, robustness, extensibility
 - Automatic processing
- Computational Thinking



Think like a scientist

- Why?

- Need for advanced analytics
- Investigation of motivations
- Looking for patterns
- Aim to understand the why

How?

- Follow the scientific method
- Research to understand the problem
- Hypothesize and come up with possible solutions
- Hypothesis testing
- Evaluation, measure results
- Critical Thinking

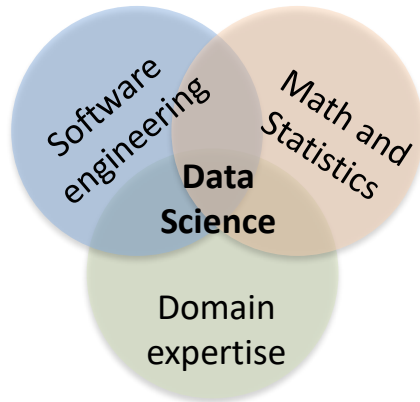


Think like a stakeholder

- Why
 - Keep business / scientific goal in mind
 - Measure performance
 - Present insights
 - Consider ethical or societal aspects
- How
 - Develop, quantify, measure, and track key performance indicators (KPIs) wrt goals
 - Presentation, communication, visualization, storytelling
 - Compliance and accountability



Summary of cross-functional thinking skills



Data Science Thinking combines ways of thinking stemming from the different disciplines it is based on.

Critical Thinking

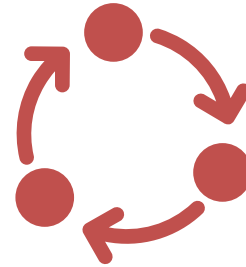
Computational Thinking

Impact orientation

Core concepts



Data Science Thinking **combines and adapts** existing ways of thinking to answer data science questions.



The process underlying Data Science Thinking is composed of **abstract steps to solve a problem of interest** to some stakeholder.

Running example



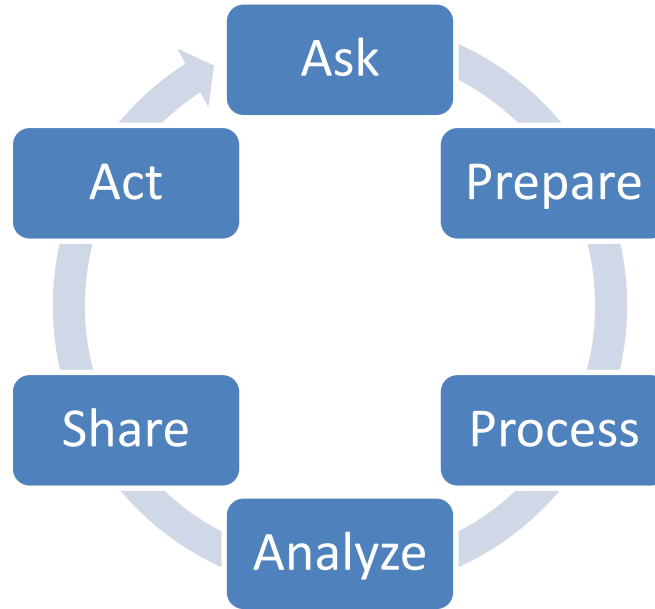
How can my athletes
get better?

I may be able to help
you based on data.
Let's see ...



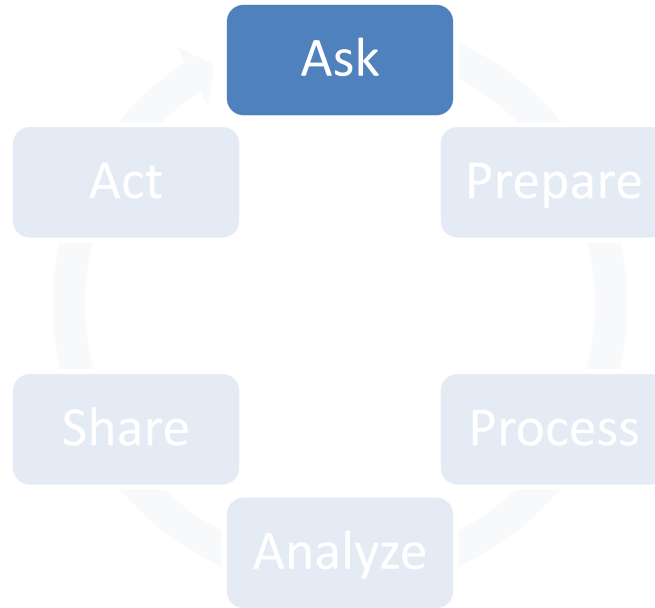


Data Science Thinking Process





Data Science Thinking Process





Ask

- **Define the problem**

- Clearly articulate the business question or problem to be solved with data.
- Identify the kind of problem.
- Understand stakeholder expectations.

- **Identify the goal**

- Determine what should be achieved by the analysis.
- Understand why it is relevant to the stakeholder(s).

Ask (Example)



How can my athletes get better?

Let's start with my sprint athletes. I want them to run faster.

You are right. Let's focus on the 100m sprint.

My athletes are world class, I expect them to run < 10s.

I may be able to help you based on data.
Let's see ...



Can you be more specific?
What kind of athletes? Better than what? In doing what? ...

The solutions probably depend on the distance to run. Which one do you have in mind?

Faster is always better of course. But what level are we talking about? What performance would you expect?

Ask (Example)



Yes. The women's team is doing great already.

My athletes all have their individual needs and styles. So I'd prefer understanding key factors to optimize their individual plan myself.



OK, I see. Given these numbers, I suppose your athletes are male, right?

Are you more interested in understanding factors playing a part in such high sprint performance or do you want recommendations for specific measures?

OK, great. I should be able to support this based on data.



Ask (Example)

- **Define the problem**
 - *Problem to be solved*: Identify key factors that play a role in male sprinters to run 100m in less than 10s.
 - *Underlying kind of problem*: Prediction of athlete performance depending on a variety of factors.
 - *Stakeholder expectations*: Actionable insights to get male team to world class level.
- **Identify the goal**
 - *Goal of the analysis*: Identified key factors may guide the definition of measures implemented to improve athlete performance.
 - *Relevance to stakeholders*: They want to make well-informed adaptations to the training plans to become internationally competitive.



Data Science Thinking Process





Prepare

- **Data collection:** Gather relevant data from various sources, such as databases, spreadsheets, or APIs.
- **Data understanding:** understand the data and the metrics you will need for analysis to assess the usefulness of the data.
- **Data access:** Evaluate and handle different data locations (internal or external), copyright, privacy, and security.
- **Data modeling:** organize and transform the data into a format that is suitable for further processing.
- **Data integration:** Combine data from different sources into a unified dataset.

Prepare (example)

Observation: Abundance of available data and factors to consider

- Performance may be influenced by diet, spikes, weather, speed, ...
- Many sources with data on many factors (historical records, measurements, logs, videos, ...)



Prepare (example)

Preparation: Collect and understand data to gather data that is **relevant** to gain insights that are actionable to adapt training plans.

- We cannot change the weather, but it is likely a relevant factor to be considered when predicting performance → Integrate wind data, temperature
- Average speed over 100m may be a too coarse granularity.
→ Focus on sources with speed measured at different times of the race
- Is it ok to consider race a factor?
 - Not actionable for the specific goal of our example → ignore
 - May be actionable for related questions, e.g., team selection. But is it ethical?

Prepare (example)

Observation: differences in data and governing policies

- Structured records (historical records), time series (measurements), recordings of races (videos), ...
- Different copyrights (open historical data, proprietary videos), privacy of personal data (blood pressure measurements, medical data), ...



Prepare (example)

Preparation: Access, model, and integrate relevant data to ultimately obtain data that are both **relevant** and **usable** to reach the analytical goal.



Access

- Download CSV and Excel files (free)

- Query time series data via APIs (restricted number of accesses)

- Stream high quality video footage (pay per query)

- Gather health data from athletes via various apps and wearables (secure access and storage)



Model

- Store all data in tables

- Avoid redundancy by uniquely representing each athlete and race



Integrate

- Correctly align race data from CSV records with health data of individual athletes

- Connect time series details with correct races

- Link videos to events and extract relevant data from videos (to be put in tables)



Prepare (example)

ID	Athlete	dateTime	result	wind
1	Tom	01:01:19:10:00	9.98	+1.2
2	Lin	12:03:14:16:00	10.21	0.00
3	Paulo	14/03/14 15:00	10.1	-0.04
4	Karl	17/05/12 08:30	10.4	+0.25

Correct
alginment

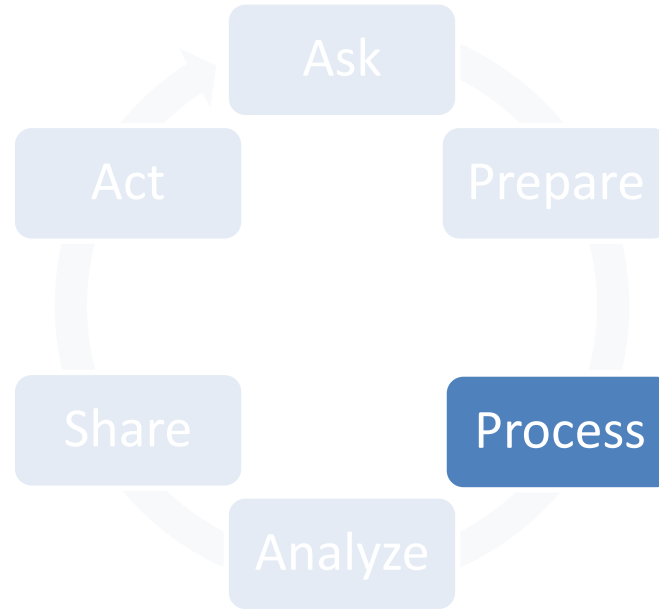
SplitID	RID	Time	Distance
1	1	2.04	10
2	1	3.14	20
3	1	3.95	30
4	1	4.78	40
5	1	5.69	50
...	...	...	...

1	2	10.21	100
1	4	2801	10
2	4	4120	20
...	...	...	...

- From downloaded CSV
- From downloaded Excel
- Extracted from video
- Retrieved via API



Data Science Thinking Process





Process

- **Data transformation:** Convert data into a suitable format for analysis, such as numerical or categorical.
- **Data cleaning:** Cleanse the data by removing errors, inconsistencies, and missing values. Necessary to avoid garbage-in-garbage-out problem.
- **Feature engineering:** Create new features or variables from existing data for analysis.



Process (example)

<u>ID</u>	Athlete	dateTime	result	wind
1	Tom	01:01:19:10:00	9.98	+1.2
2	Lin	12:03:14:16:00	10.21	0.00
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<u>SplitID</u>	<u>RID</u>	Time	Distance
1	1	2.04	10
2	1	3.14	20
3	1	3.80	30
4	1	4.20	40
5	1	5.18	50
...	...	...	...
1	2	10.21	100
1	4	2801	10
2	4	4120	20
...	...	...	...



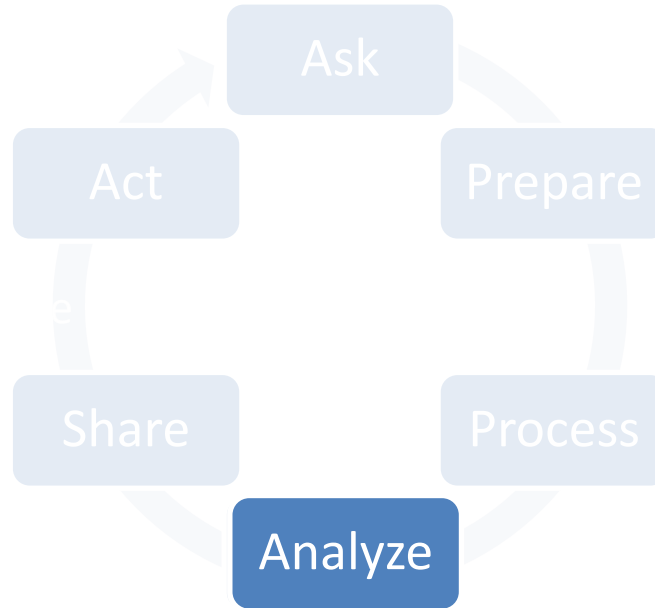
Process (example)

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3	Paulo	14:03:14:15:00	10.13	-0.04
4	Karl	17:05:12:08:30	10.42	+0.25

<u>SplitID</u>	<u>RID</u>	Time	Distance	Velocity
1	1	2.04	10	4.90
2	1	3.14	20	9.09
3	1	3.95	30	12.35
4	1	4.78	40	12.04
5	1	5.69	50	10.98
...	...	...	...	...
1	2	10.21	100	
1	4	2.80	10	3.57
2	4	4.12	20	7.58
...	...	...	...	...



Data Science Thinking Process



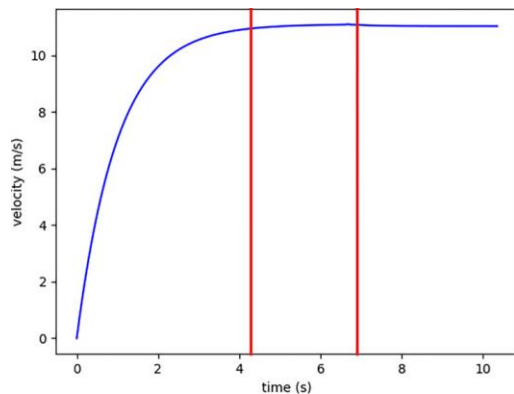


Analyze

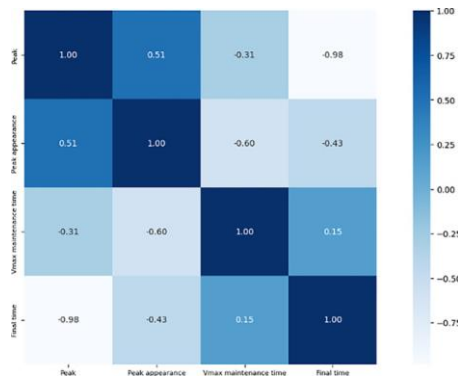
- **Exploratory data analysis:** Explore the data to understand its characteristics and identify patterns.
- **Statistical analysis:** Apply statistical techniques to test hypotheses and draw inferences.
- **Machine learning:** Utilize machine learning algorithms to uncover hidden insights and make predictions.



Analyze (example)



Through exploratory analysis, recognize the pattern that sprints typically divided into three phases (acceleration, peak, deceleration)



Study correlations of factors or metrics using statistical tests to determine that length of maximum velocity is not strongly correlated with final time.

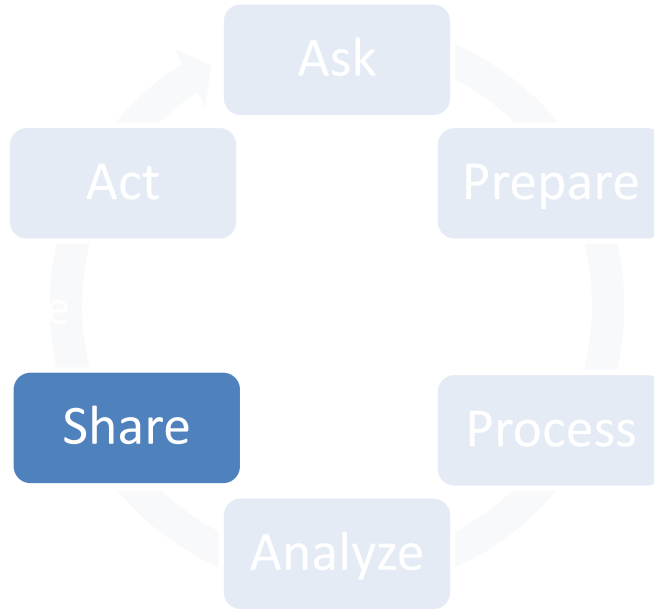


Train a predictive model using machine learning to predict realistic performance under various conditions.

Images from <https://doi.org/10.1371/journal.pone.0303366>



Data Science Thinking Process





Share

- **Data visualization:** Create visualizations (charts, graphs, dashboards) to communicate findings effectively.
- **Storytelling:** Present the findings in a clear and concise manner, tailored to the audience's needs.
- **Reporting:** Generate reports summarizing the key findings and recommendations.

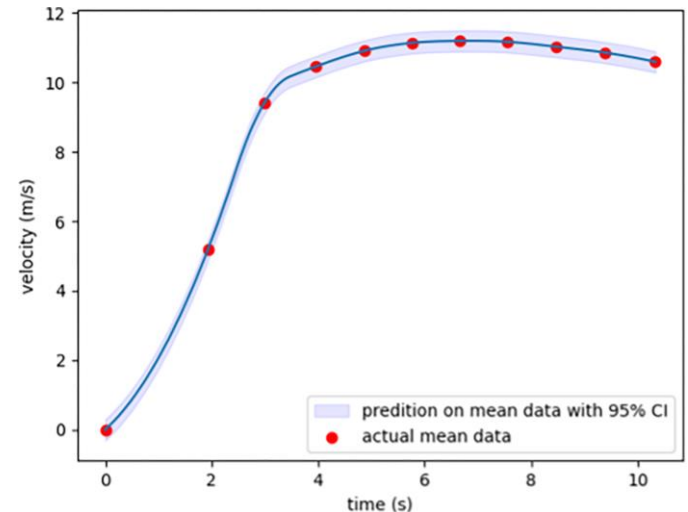
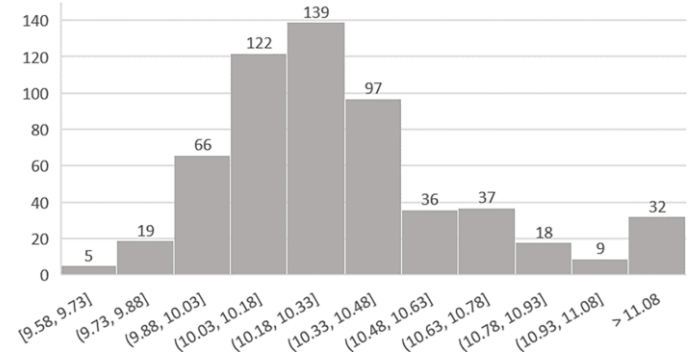
Share (Example)

Our model can accurately predict velocity over 10s splits (see bottom graph), as tested on a large variety of races in terms of final time (see histogram). You can use this to see at which phase your athletes have the most room for improvement.

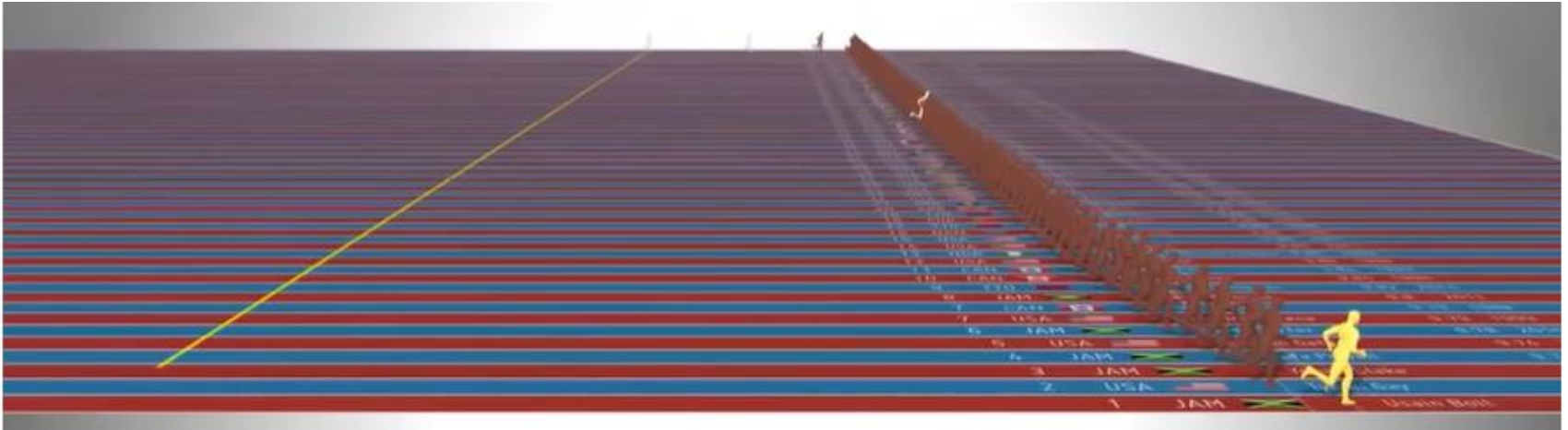
Peak velocity shows by far the strongest negative correlation with final time. This may help prioritize training increasing peak velocity.



Histogram of final time



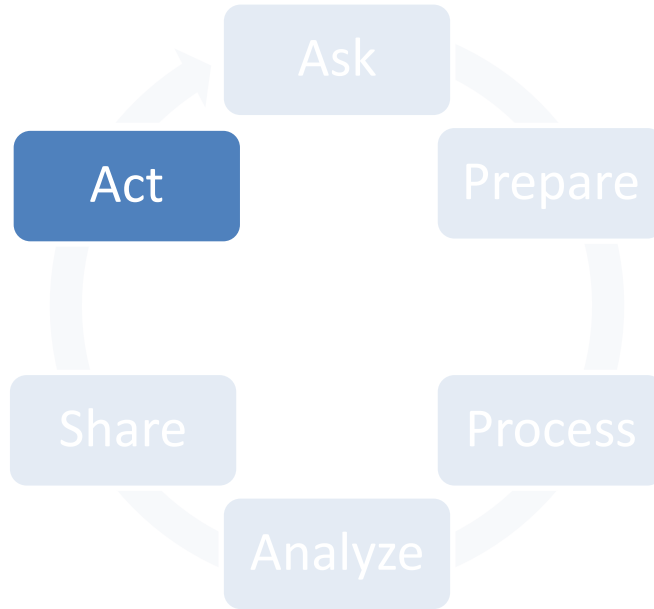
Share (example)



[Facebook20]



Data Science Thinking Process





Act

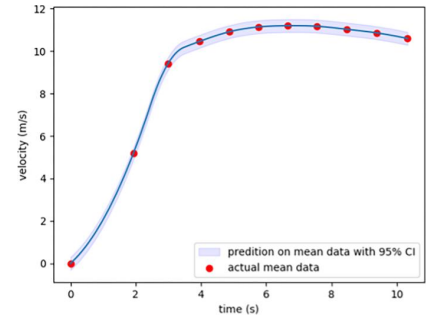
- **Decision-making:** Use the insights gained from the analysis to make informed decisions.
- **Implementation:** Put the decisions into action to achieve the desired outcomes.
- **Tracking:** Define and implement measures and metrics to monitor impact of the implementation.

Act (Example)



Tom is doing exceptionally well in the first half of the race, but performance deteriorates in second half. Let's work on reducing deceleration.

Let's implement the following measures to that end: 🏃💓🏃



<u>ID</u>	Athlete	dateTime	result	wind
1	Tom	01:01:19:10:00	9.98	+1.2

<u>SplitID</u>	<u>RID</u>	Time	Distance	Velocity
1	1	2.04	10	4.90
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3	1	3.95	30	12.35
4	1	4.78	40	12.04
5	1	5.69	50	10.98



Data Science Thinking Process



Ask (iterative)

- **Reflect on results:** Evaluate the effectiveness of the analysis and the actions taken.
- **Identify new questions:** Based on the insights gained, formulate new questions for further exploration.
- **Start the process again:** Begin a new cycle of data analysis to address the new questions.

Act (Example)

... as time has passed...



Wow, Tom made great progress! Most of my other athletes, too.

Can we find out which measures I tried were most effective to optimize future training plans?

I may be able to help you based on data.
Let's see ...



<u>ID</u>	Athlete	dateTime	result	wind
1	Tom	01:01:24:10:00	9.79	+0.9

<u>SplitID</u>	<u>RID</u>	Time	Distance	Velocity
1	1	2.02	10	4.93
2	1	3.12	20	9.12
3	1	3.94	30	12.25
4	1	4.76	40	12.23
5	1	5.66	50	11.99



Summary

- Make decisions
- Implement actions
- Monitor KPIs

- Visualize
- Report
- Captivate stakeholders



- Define the problem
- Identify the goals

- Collect data
- Understand data
- Access data
- Model data
- Integrate data

- Transform data
- Clean data
- Enrich data

- Use exploratory analysis
- Apply statistical methods
- Leverage machine learning

Computational Thinking and Data Science Thinking

COMPARISON



Similarities

Both Computational Thinking and Data Science Thinking ...

- ... define thought processes that help solve complex problems.
- ... involve problem solving and analytical skills.
- ... leverage computational tools.

Differences

	Computational Thinking	Data Science Thinking
Focus	Broad, general problem-solving	Focus on gaining data-based insights
Domain-knowledge	Not strictly required	Required
Tools and techniques	Wide range of computational tools and techniques	Focus on data-oriented tools and techniques, scientific methodology
Outcome	Efficient and systematic solutions	Data-based insights

Computational Thinking and Data Science Thinking

SUMMARY



Summary (data science perspective)

**Computational
Thinking**

is part of

**Data Science
Thinking**

follows

**Process for
data-based insights**



Decomposition



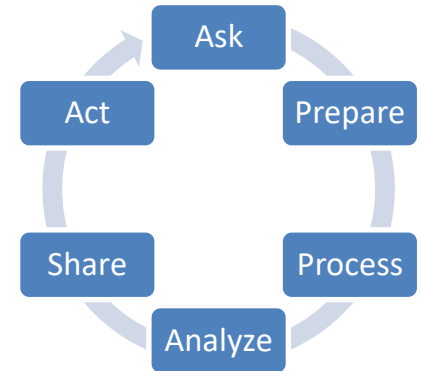
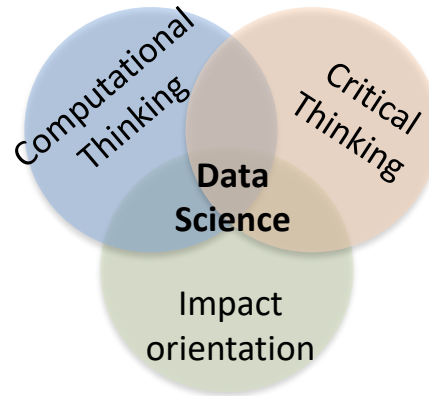
Pattern recognition



Abstraction

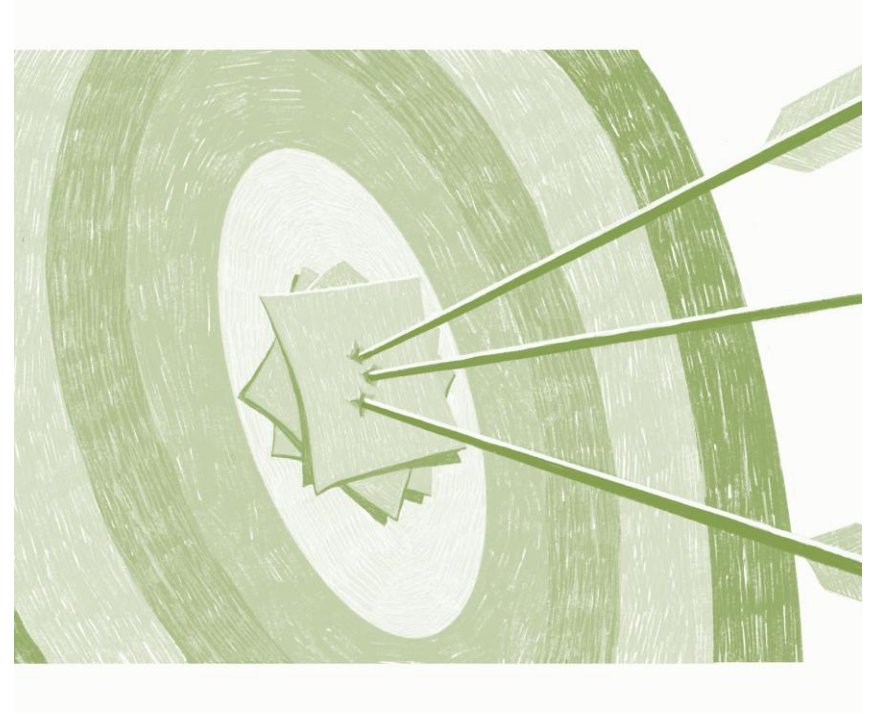


Algorithm design



You are now equipped to ...

- ...describe and compare the main characteristics of Computational Thinking and Data Science Thinking.
- ... explain how cross-functional thinking skills complement each other to solve a data science problem.
- ... argue why and how a problem should be addressed through Data Science Thinking (or not).
- ... conceive a rough agenda to solve a new problem by following the Data Science Thinking process.



Questions?



References / Additional reading

- [Facebook20] <https://www.facebook.com/EMsportscience/videos/from-1896-to-our-days-the-evolution-of-100m-sprint/866569130746512/>
- [EMC14] EMC Education Services, ed. *Data science and big data analytics: discovering, analyzing, visualizing and presenting data*. John Wiley & Sons, 2014.